Astrostatistics – Activity 1

Descriptive Statistics, Outliers, Gaussianity, Correlation, and Conditional Probability

Context and goals

In this activity, you will work with a large, real astronomical dataset from the NASA Exoplanet Archive. The objective is not to perform a detailed study of exoplanet physics, but to use a real observational catalog to practice fundamental statistical concepts that appear throughout astrophysics.

Dataset

Use the **Planetary Systems Composite Parameters** table (`pscomppars`) from the NASA Exoplanet Archive.

At minimum, download the following columns:

Column	Meaning
`pl_name`	Planet name
`discoverymethod`	Discovery method
`disc_year`	Discovery year
`pl_orbper`	Orbital period [days]
`pl_rade`	Planet radius [$R_\oplus$]
`pl_bmasse`	Best available planet mass estimate [$M_\oplus$]
`pl_bmassprov`	Provenance of the mass estimate
`pl_orbeccen`	Orbital eccentricity

You may download the table through the Archive interface or through its TAP service. A suitable ADQL query is

```
SELECT pl_name, discoverymethod, disc_year, pl_orbper,
       pl_rade, pl_bmasse, pl_bmassprov, pl_orbeccen
FROM pscomppars
```

Record the **date on which you downloaded the catalog**. The NASA Exoplanet Archive is continuously updated, so the exact number of objects may change with time.

Important: Do not remove every row containing a missing value at the beginning of the analysis. Different parts of the activity require different variables. Instead, remove missing or invalid values only for the variables needed in each section.

1 Initial exploration of the catalog

Before calculating any statistical quantity, inspect the dataset.

1. How many planets are present in the downloaded catalog?
2. For each variable used in this activity, determine the number and fraction of missing values.
3. How many different discovery methods are represented?
4. What is the most common discovery method? In this case, the statistical mode has a natural interpretation because `discoverymethod` is a categorical variable.
5. Briefly discuss why missing values should not automatically be interpreted as failed observations.

2 Descriptive statistics of the orbital period

For this section, select all planets with a finite and positive orbital period,

$$P \equiv \texttt{pl_orbper} > 0.$$

Let the resulting sample contain N planets.

2.1 Measures of location and scale

Calculate:

- sample size N;
- arithmetic mean;
- median;
- sample variance;
- sample standard deviation;
- skewness;
- excess kurtosis.

For the kurtosis, use the convention in which a Gaussian distribution has

$$K_{\text{excess}} = 0.$$

2.2 What does "mode" mean for a continuous variable?

The orbital period is a continuous quantity. Therefore, an exact numerical mode may depend on measurement precision and repeated catalog values.

Instead of simply applying a function that returns the most frequently repeated floating-point number:

(a) construct a histogram of P;

(b) identify the **modal interval**, i.e. the histogram bin containing the largest number of objects;

(c) repeat the histogram using at least two reasonable choices of bin width;

(d) discuss whether the estimated mode changes.

What does this tell you about the meaning of the mode for a continuous distribution?

2.3 First visualization

Construct:

(a) a histogram of P using a linear horizontal axis;

(b) a histogram of $\log_{10} P$.

Compare the two figures. Which one reveals the structure of the dataset more clearly? Why?

3 Quartiles and percentiles

Using the orbital-period sample, calculate

$$P_5, \quad P_{10}, \quad P_{16}, \quad P_{25}, \quad P_{50}, \quad P_{75}, \quad P_{84}, \quad P_{90}, \quad P_{95}.$$

Identify the quartiles

$$Q_1 = P_{25}, \qquad Q_2 = P_{50}, \qquad Q_3 = P_{75},$$

and calculate the interquartile range

$$IQR = Q_3 - Q_1.$$

Answer the following questions:

1. What fraction of the observations lies between Q_1 and Q_3?

2. What fraction lies between P_{16} and P_{84}?

3. Why are P_{16} and P_{84} often useful in astronomy when comparing a distribution with a Gaussian distribution?

4. Compare the median with the mean. Is the orbital-period distribution symmetric?

3.1 Boxplot

Construct a boxplot of P and identify:

- Q_1;
- the median;
- Q_3;
- the interquartile range;
- observations flagged as possible outliers.

Because the period distribution spans a large dynamic range, also construct a boxplot for $\log_{10} P$. Compare the information provided by the histogram and the boxplot.

4 Outliers and robustness

Use Tukey's rule to flag possible outliers in the **linear orbital-period variable**:

$$P < Q_1 - 1.5\,IQR$$

or

$$P > Q_3 + 1.5\,IQR.$$

1. Determine the lower and upper Tukey limits.
2. How many planets are flagged as outliers?
3. What fraction of the orbital-period sample do they represent?
4. Construct a new sample after removing the flagged observations.
5. Recalculate the mean, median, standard deviation, skewness, and excess kurtosis.
6. Compare these statistics before and after outlier removal.

Which statistics are the most sensitive to the removal of extreme observations? Which are the most robust?

Interpretation question: Does being identified as a statistical outlier imply that an observation is physically incorrect or should be removed from a scientific analysis? Explain.

5 Does the orbital-period distribution follow a Gaussian distribution?

Use the sample obtained after the Tukey outlier removal.

5.1 Linear orbital period

For P:

(a) construct a normalized histogram;

(b) calculate the sample mean μ and standard deviation σ;

(c) overplot a Gaussian distribution with the same μ and σ;

(d) construct a Q–Q plot against a Gaussian distribution;

Use

$$H_0: \quad \text{the data are drawn from a Gaussian distribution}$$

and

$$H_1: \quad \text{the data are not drawn from a Gaussian distribution.}$$

Adopt a significance level

$$\alpha = 0.05.$$

State the resulting p-value and your statistical decision.

5.2 Logarithmic orbital period

Now define

$$Y = \log_{10} P.$$

Using the **same planets retained after the outlier cut**, repeat the Gaussianity analysis for Y:

- histogram + fitted Gaussian;
- Q–Q plot;
- skewness;
- excess kurtosis;

Then answer:

1. Does the histogram appear symmetric and bell-shaped, or does it show skewness, heavy tails, multiple peaks, or other deviations from a Gaussian shape?
2. In the Q–Q plot, do the points approximately follow a straight line? If deviations are present, do they occur mainly in the center of the distribution or in the tails?
3. Is the skewness close to zero? What does its sign and magnitude indicate about the asymmetry of the distribution?

4. Is the excess kurtosis close to zero? Does its value suggest heavier or lighter tails than a Gaussian distribution?

5. Considering all of these diagnostics together, do they lead to the same conclusion about Gaussianity? If not, explain which diagnostics disagree and why this may occur.

6 Conditional probability and selection effects

Return to the sample of planets with a valid orbital period.

Define the events

$$T = \{\text{planet was discovered by the Transit method}\}$$

and

$$S = \{P < 10 \text{ days}\}.$$

Use observed frequencies in the catalog as empirical estimates of probabilities.

6.1 Probabilities

Calculate

$$\mathsf{P}(T), \qquad \mathsf{P}(S), \qquad \mathsf{P}(T \cap S),$$

$$\mathsf{P}(S \mid T), \qquad \mathsf{P}(S \mid T^c),$$

and

$$\mathsf{P}(T \mid S).$$

Verify directly from the catalog that

$$\mathsf{P}(S \mid T) = \frac{\mathsf{P}(S \cap T)}{\mathsf{P}(T)}.$$

6.2 Interpretation

Answer:

1. Is $\mathsf{P}(S \mid T)$ equal to $\mathsf{P}(S)$?

2. Based on the empirical frequencies, do S and T appear independent?

3. Is $\mathsf{P}(S \mid T)$ equal to $\mathsf{P}(T \mid S)$? Explain why these quantities have different meanings.

4. What observational or selection effect could help explain a high fraction of short-period planets among transit discoveries?

Suggested submission

Submit a Jupyter notebook or equivalent reproducible analysis containing:

- all data selections and calculations;
- clearly labeled figures with physical units;
- the numerical answers requested above;
- short interpretations accompanying the main results;
- the final synthesis discussion.

Your code should reproduce all reported numerical results from the original downloaded catalog. Do not manually enter numerical answers that can be generated from the analysis.

Data source

NASA Exoplanet Archive, Planetary Systems Composite Parameters (`pscomppars`). Column definitions and TAP documentation are available through the NASA Exoplanet Archive documentation pages.